Simone Carletti

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About Me

I am a **Computer Engineering M.Sc.** student specializing in **Robotics and Autonomous Systems**. I'm currently completing my thesis at **EPFL**, where I am developing a control system for a morphing-wing drone using $\mathcal{L}_1$ **Adaptive Control** and **Reinforcement Learning**. I have hands-on experience across the robotics stack, from **perception** and **planning** to **control systems**, working with **ROS2** and **embedded systems**. I'm proficient in **C++**, **Python**, and **TypeScript**, with research experience at EPFL and DTU. Outside the lab, I enjoy dancing salsa, hiking, taking pictures, and exploring new music.

Education

EPFL - LIS, *Master's Thesis* — *Sept 2025 – Present*

- Title: L1-Adaptive augmentation for robust morphing-wing drone flight control
- Skills: Control Systems, RL (PPO), C++, Python, Simulink, ROS2, Git, Linux, Unity, PX4
- Validated the algorithms through hardware-in-the-loop simulations and real flights

DTU, *M.Sc. Autonomous Systems* — *Sept 2024 – Jul 2025*

- Erasmus+ exchange - Converted GPA: 30/30

Politecnico di Torino, *M.Sc. Computer Engineering* — *Sept 2023 – Present*

- Curriculum in Automation and Intelligent Cyber-Physical Systems - Final GPA: 29.48/30

Università degli Studi di Firenze, *B.Sc. Computer Science and Engineering* — *Sept 2020 – Sept 2023*

- Graduated with 110/110 cum laude (with honors)
- Thesis: Sensor network to monitor the greenhouses of the Botanical Garden of Florence

Experience

Robotics Engineer, *DIANA student team* — *Sept 2023 – Jul 2024*, *Turin, Italy*

- Collaborated on a Mars rover student project, implementing perception and EKF state estimation in C++ while contributing to software development, testing, and system integration

Full-Stack Developer, *Genomedics S.r.l* — *Sept 2020 – Present*, *Florence, Italy*

- Lead developer of three key projects that handle the release, deployment, and licence management for all the company's products. Developer for the company's main project (KPlatform)
- Skills: Angular (Typescript, SCSS, HTML), Vue.Js, PHP, Laravel, SQL, Python, Git, Linux

Selected Projects

Autonomous Navigation & Task Execution - DTU RoboCup — *Feb 2025 – May 2025*

- Developed autonomous navigation, control, and task execution algorithms with continuous integration and validation tests on the real platform

Digital Control - Autonomous Robotics — *Oct 2024 – Dec 2024*

- Designed digital controllers and implemented them on embedded hardware (GitHub)

3D Object Tracking using Stereo Camera — *Nov 2024 – Dec 2024*

- Developed a 3D perception pipeline integrating Faster R-CNN for detection, SIFT-based stereo imaging, and EKF for robust object tracking (GitHub)

Comparison and implementation (from scratch) of YOLOv1 and U-Net — *Oct 2024 – Dec 2024*

- Implemented landmark computer vision models from scratch using PyTorch (GitHub)

Localization with EKF in ROS2 — *Nov 2023 – Dec 2023*

- Implemented a full EKF localization pipeline from scratch in ROS2 and Python, fusing data from LiDAR, IMU, and wheel odometry

Skills & Activities

- **Languages**: Fluent in English, Native Italian speaker, Basic Spanish skills
- **Programming**: C++, C, Python, TypeScript, JavaScript, Rust, Java, Matlab, PHP, SQL, HTML, CSS/SCSS, Assembly, LaTeX
- **Tools**: PyTorch, Linux, Git, Google Colab, OpenCV, Onshape, CMake, Docker, Node-RED, Multisim
- **Other**: Raspberry Pi, Arduino, ESP8266, NodeMCU, Soldering, Nvidia, Jetson/Orin, Guitar, Photography, Networking
- **Volunteering**: ESN Copenhagen and DTU (Organized events, supported international students and promoted cultural exchange)